

ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis

St. Louis, MO

Bachelor of Science in Mechanical Engineering

Expected May 2027

GPA: 3.65 / 4.00

- Relevant Coursework: Flight Dynamics; Aircraft Design; Modeling, Simulation, and Control; Vibrations; Numerical Methods and Matrix Algebra; Fluid Mechanics;

TECHNICAL SKILLS

Programming & Modeling: MATLAB/Simulink, Python, C++, Git

Controls & Estimation: PID control, sensor fusion, EKF, IMU calibration, state estimation

Hardware & Design Tools: Pixhawk/Cube, ArduPilot, Mission Planner, MAVLink, DroneCAN, HIL testing, flight-log analysis, SolidWorks (GD&T), ANSYS, XFLR5, Teensy/Arduino, composites

RESEARCH EXPERIENCE

UAS Research Intern, Missouri Institute for Defense & Energy, Kansas City, MO

May 2026 – Aug. 2026

Supervisor: Dr. Travis Fields, Deputy Director.

- Developed an experimental unmanned aerial vehicle (UAV) testbed of 7 quadcopter platforms from component-level hardware (Cube Black, ArduPilot, sensors, propulsion, power), defining a repeatable methodology for configuration control, sensor calibration, and bench validation.
- Investigated how payload placement and nose mass affect center-of-gravity location and static stability margin using ArduPilot flight logs and MATLAB center-of-gravity models on Tigrar unmanned aircraft system (UAS) platforms; supported flight-test campaigns after composite structural repair.
- Characterized the attitude-estimation-and-control loop (measure–estimate–command–actuate) on Cube Black/ArduPilot, using Python/MAVLink and DroneCAN to compare live attitude estimates with physical motion and commanded servo response.

Sensor Fusion Research Intern, MIRO Lab, UMKC, Kansas City, MO

May 2025 – Aug. 2025

Advisor: Dr. Hee-Sup Shin.

- Built a Teensy 4.0 multimodal sensing platform synchronizing dual EMG and dual IMU streams at 1 kHz and conducted human-subject trials across rest, MVC, and arm-motion conditions.
- Implemented MATLAB EKF calibration and visualization fusing IMU and visual ground-truth measurements to reduce drift in motion estimates; contributed to an IEEE Body Sensor Networks 2025 abstract and poster, *Toward Electromechanical Measurement of Muscle Activity*.

ENGINEERING EXPERIENCE

Lead Avionics Engineer, WashU VTOL, St. Louis, MO

Sept. 2025 – Present

- Defined avionics architecture for a prototype hex VTOL (Pixhawk 6C, sensors, communications, isolated 4S avionics/6S propulsion power) and configured VTOL/fixed-wing modes, PID gains, and failsafes.
- Diagnosed motor-induced EMI causing I2C barometer/magnetometer dropouts and restored reliable telemetry via rerouting, twisted-pair I2C, and pull-up resistors.
- Validated the integrated avionics stack through bench testing, HIL simulation, first hover, and VTOL-to-fixed-wing transition.

Payload Systems Engineer, WashU Design/Build/Fly, St. Louis, MO

Sept. 2024 – Present

Structures – Aerodynamics – Payload Systems; returned Fall 2026 (away Spring 2026).

- Sized carbon-fiber wing spars under 2.5 g loads (ANSYS FEA), designed payload-release hardware with ASME Y14.5 GD&T, compared XFLR5/ANSYS CFD with flight telemetry, and now lead Mission 3 towed-sensor payload systems for the AIAA Design/Build/Fly 2026–2027 competition.

LEADERSHIP & COMMUNICATION EXPERIENCE

Teaching Assistant, Washington University in St. Louis, St. Louis, MO

Aug. 2025 – Present

Grade assignments and hold office hours for Vibrations (Aug. 2026 – Present) and MEMS 2000 Circuits (Aug. 2025 – Dec. 2025).